

Quantum-to-classical Lax correspondence for Class 5 and Class 6 models

1 Introduction

2 Quantum lattice setup and continuum limit

2.1 Quantum spin chains and the lattice Lax equation

The two lattice models studied in this paper are homogeneous periodic spin chains with nearest-neighbour interactions and a two-dimensional local Hilbert space. We use the standard quantum inverse-scattering framework for such chains [5, 6], and first fix the common notation before introducing any model-dependent Hamiltonian or R -matrix.

Let $\mathcal{H}_{\text{loc}} \simeq \mathbb{C}^2$ denote a reference local Hilbert space. Consider a chain with N lattice sites, and let $\mathcal{H}_n$ be a copy of $\mathcal{H}_{\text{loc}}$ assigned to the n th site,

$$\mathcal{H}_n \simeq \mathcal{H}_{\text{loc}} \simeq \mathbb{C}^2, \quad n = 1, \dots, N. \quad (2.1)$$

The Hilbert space of the full chain is the tensor product

$$\mathcal{H}^{(N)} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \dots \otimes \mathcal{H}_N. \quad (2.2)$$

We impose periodic boundary conditions, so the site labels are understood modulo N ; in particular, the site following N is site 1.

A homogeneous nearest-neighbour Hamiltonian is specified by a single *two-site Hamiltonian operator*

$$h \in \text{End}(\mathcal{H}_{\text{loc}} \otimes \mathcal{H}_{\text{loc}}). \quad (2.3)$$

The adjective “homogeneous” means that the same two-site operator h is used on every nearest-neighbour bond. We denote by $h_{n,n+1}$ the operator on the full Hilbert space (2.2) that acts as h on $\mathcal{H}_n \otimes \mathcal{H}_{n+1}$ and as the identity on all remaining sites. For $1 \leq n < N$, this embedding is explicitly

$$h_{n,n+1} = \mathbf{1}_{\mathcal{H}_1} \otimes \dots \otimes \mathbf{1}_{\mathcal{H}_{n-1}} \otimes h \otimes \mathbf{1}_{\mathcal{H}_{n+2}} \otimes \dots \otimes \mathbf{1}_{\mathcal{H}_N}. \quad (2.4)$$

For the bond joining the last and first sites, $h_{N,1}$ denotes the same operator h acting on $\mathcal{H}_N \otimes \mathcal{H}_1$, with the other sites left unchanged. Thus $h_{n,n+1}$ is the copy of the fixed two-site operator h on the n th bond.

With these conventions, the Hamiltonian of the periodic chain is

$$H^{\text{lat}} = \sum_{n=1}^N h_{n,n+1}, \quad h_{N,N+1} \equiv h_{N,1}. \quad (2.5)$$

The model dependence therefore enters through the choice of the single matrix h . We next introduce the quantum R -matrix and the lattice Lax equation associated with a nearest-neighbour integrable chain.

For the integrable chains considered in this paper, the local interaction is encoded by a quantum R -matrix. Introduce an auxiliary space

$$\mathcal{V}_a \simeq \mathcal{H}_{\text{loc}} \simeq \mathbb{C}^2, \quad (2.6)$$

which is a tensor factor distinct from the physical Hilbert spaces $\mathcal{H}_n$. When several auxiliary spaces are required, we label them by a, b, c .

Let $u \in \mathbb{C}$ denote the *quantum spectral parameter*. A quantum R -matrix in the representation used here is a family of two-site operators

$$R(u) \in \text{End}(\mathcal{H}_{\text{loc}} \otimes \mathcal{H}_{\text{loc}}). \quad (2.7)$$

Whenever two tensor factors are individually labelled, the corresponding subscripts indicate the copy of this same two-site operator acting on those factors. Thus, for example, $R_{ab}(u) \in \text{End}(\mathcal{V}_a \otimes \mathcal{V}_b)$ and $R_{a,n}(u) \in \text{End}(\mathcal{V}_a \otimes \mathcal{H}_n)$.

The models considered here have an R -matrix of difference form. Its Yang–Baxter equation is

$$\begin{aligned} R_{ab}(u-v) R_{ac}(u-w) R_{bc}(v-w) \\ = R_{bc}(v-w) R_{ac}(u-w) R_{ab}(u-v), \end{aligned} \quad (2.8)$$

which is an operator identity on $\mathcal{V}_a \otimes \mathcal{V}_b \otimes \mathcal{V}_c$ [5]. The subscripts indicate the two tensor factors on which each copy of R acts.

Let

$$P \in \text{End}(\mathcal{H}_{\text{loc}} \otimes \mathcal{H}_{\text{loc}}) \quad (2.9)$$

be the permutation operator defined by $P(|v\rangle \otimes |w\rangle) = |w\rangle \otimes |v\rangle$. As for the R -matrix, P_{ab} denotes its copy on $\mathcal{V}_a \otimes \mathcal{V}_b$. We normalize the R -matrix so that its regular point is $u = 0$,

$$R(0) = P, \quad \text{equivalently} \quad R_{ab}(0) = P_{ab}. \quad (2.10)$$

For a regular fundamental spin- $\frac{1}{2}$ chain, the two-site Hamiltonian operator introduced in (2.3) is obtained from the same two-site R -matrix as

$$h = P \left. \frac{\partial R(u)}{\partial u} \right|_{u=0}. \quad (2.11)$$

This is the standard relation between a regular R -matrix and the nearest-neighbour Hamiltonian density in the quantum inverse-scattering construction [5, 1].

A scalar rescaling $R(u) \mapsto f(u)R(u)$ with $f(0) = 1$ preserves the regularity convention (2.10) and adds $f'(0)\mathbf{1}$ to h . This term shifts the total energy by a constant and leaves the Heisenberg equations unchanged. An overall rescaling of the spectral parameter amounts to a corresponding normalization of the Hamiltonian. These normalization freedoms are fixed together with each explicit R -matrix.

To associate the R -matrix with a particular physical site, one copy of its second tensor factor is identified with the local Hilbert space $\mathcal{H}_n$. The resulting *quantum lattice Lax operator* is

$$L_{a,n}(u) := R_{a,n}(u) \in \text{End}(\mathcal{V}_a \otimes \mathcal{H}_n). \quad (2.12)$$

For a fixed physical site n , the Yang–Baxter equation implies the local exchange relation

$$R_{ab}(u-v) L_{a,n}(u) L_{b,n}(v) = L_{b,n}(v) L_{a,n}(u) R_{ab}(u-v), \quad (2.13)$$

acting on $\mathcal{V}_a \otimes \mathcal{V}_b \otimes \mathcal{H}_n$.

The ordered product of lattice Lax operators around the periodic chain defines the *quantum monodromy matrix*

$$T_a(u) := L_{a,N}(u)L_{a,N-1}(u) \cdots L_{a,1}(u), \quad (2.14)$$

and tracing over the auxiliary space gives the *transfer matrix*

$$\tau(u) := \text{tr}_a T_a(u). \quad (2.15)$$

Equation (2.13) implies

$$[\tau(u), \tau(v)] = 0, \quad (2.16)$$

so an expansion of $\tau(u)$ generates mutually commuting quantum charges [5]. At the regular point, the logarithmic derivative of the transfer matrix reproduces the nearest-neighbour Hamiltonian (2.5), up to the scalar and normalization freedoms described after (2.11).

We next need the local equation governing the time evolution of $L_{a,n}(u)$. Differentiating the Yang–Baxter relation at its regular point and using (2.11) gives the *Sutherland relation* [1]

$$\begin{aligned} [h_{n-1,n}, L_{a,n}(u)L_{a,n-1}(u)] \\ = L_{a,n}(u) \partial_u L_{a,n-1}(u) - (\partial_u L_{a,n}(u)) L_{a,n-1}(u). \end{aligned} \quad (2.17)$$

Here $h_{n-1,n}$ denotes the embedding, defined above, of the same two-site Hamiltonian h on the neighbouring physical sites $n-1$ and n .

Let t_{lat} denote the quantum lattice time. For an operator $\mathcal{O}$ with no explicit time dependence we use the Heisenberg equation

$$\partial_{t_{\text{lat}}} \mathcal{O} = i[H^{\text{lat}}, \mathcal{O}]. \quad (2.18)$$

Because $L_{a,n}$ acts only on the physical site n , only the two Hamiltonian terms $h_{n-1,n}$ and $h_{n,n+1}$ contribute to its commutator with H^{lat} . The Sutherland relation organizes these two contributions into a discrete zero-curvature equation.

For spectral parameters at which $L_{a,n}(u)$ is invertible, define the *finite-lattice time Lax operator*

$$\begin{aligned} A_{a,n}(u) &= i h_{n-1,n} - i L_{a,n}(u)^{-1} h_{n-1,n} L_{a,n}(u) \\ &\quad - i L_{a,n}(u)^{-1} \partial_u L_{a,n}(u). \end{aligned} \quad (2.19)$$

The physical part of $A_{a,n}$ acts on the neighbouring sites $n-1$ and n , while its matrix indices belong to the auxiliary space $\mathcal{V}_a$. Equations (2.17) and (2.18) then give

$$\partial_{t_{\text{lat}}} L_{a,n}(u) = A_{a,n+1}(u) L_{a,n}(u) - L_{a,n}(u) A_{a,n}(u). \quad (2.20)$$

This is the lattice analogue of a zero-curvature equation. The time Lax operator has a scalar ambiguity: adding to every $A_{a,n}$ the same scalar function of u times the identity on the full tensor product leaves (2.20) unchanged. This freedom permits convenient normalizations.

A structural point in (2.20) enters the continuum limit. The two products on its right-hand side contain operators that share a physical lattice site: $A_{a,n+1}$ and $L_{a,n}$ both act on $\mathcal{H}_n$, while $L_{a,n}$ and $A_{a,n}$ also overlap on $\mathcal{H}_n$. Consequently, coherent-state expectation values of these products retain correlations associated with the shared physical site.

2.2 Coherent states and long-wavelength limit

We next describe the common kinematical ingredients used to pass from the spin chain to a smooth classical spin field. We use standard $SU(2)$ spin coherent states [9]; their use in long-wavelength limits of integrable spin chains is reviewed, for example, in [6]. The same construction is used in the direct continuum analysis of the Class 5 and Class 6 chains [2].

Let $\sigma^1, \sigma^2, \sigma^3$ denote the Pauli matrices on the reference local Hilbert space $\mathcal{H}_{\text{loc}} \simeq \mathbb{C}^2$, and let $|\uparrow\rangle$ and $|\downarrow\rangle$ be the eigenvectors of σ^3 with eigenvalues $+1$ and -1 , respectively. The spin- $\frac{1}{2}$ operators are normalized as

$$s^i := \frac{1}{2}\sigma^i, \quad i = 1, 2, 3. \quad (2.21)$$

Throughout, repeated Cartesian indices $i, j, k \in \{1, 2, 3\}$ are summed. At the n th lattice site, a normalized spin coherent state is parametrized by polar and azimuthal angles θ_n and ϕ_n as

$$|\Omega_n\rangle = \cos \frac{\theta_n}{2} |\uparrow\rangle + e^{i\phi_n} \sin \frac{\theta_n}{2} |\downarrow\rangle. \quad (2.22)$$

This phase convention fixes the associated unit vector. We define the *classical spin vector at site n* by

$$\mathbf{S}_n := \langle \Omega_n | \boldsymbol{\sigma} | \Omega_n \rangle = \begin{pmatrix} \sin \theta_n \cos \phi_n \\ \sin \theta_n \sin \phi_n \\ \cos \theta_n \end{pmatrix}, \quad \boldsymbol{\sigma} := (\sigma^1, \sigma^2, \sigma^3). \quad (2.23)$$

It follows directly that

$$\mathbf{S}_n^\top \mathbf{S}_n = 1, \quad \langle \Omega_n | s^i | \Omega_n \rangle = \frac{1}{2} S_n^i. \quad (2.24)$$

Thus the normalization of the classical spin field is fixed independently of the model-dependent normalization of the Lax matrix.

For a configuration of the full chain we use the normalized product coherent state

$$|\Omega\rangle := |\Omega_1\rangle \otimes \cdots \otimes |\Omega_N\rangle \in \mathcal{H}^{(N)}. \quad (2.25)$$

The coherent-state labels are allowed to depend on time. We denote by t the time coordinate of the continuum field theory. Its relation to the quantum lattice time t_{lat} introduced in (2.18) is part of the continuum scaling.

Let $\epsilon > 0$ denote the *lattice spacing*. We assign the spatial coordinate

$$x_n := (n-1)\epsilon, \quad n = 1, \dots, N, \quad (2.26)$$

and take the long-wavelength limit

$$\epsilon \rightarrow 0, \quad N \rightarrow \infty, \quad \ell := N\epsilon \text{ fixed}, \quad (2.27)$$

where ℓ is the physical circumference of the continuum spatial circle. The long-wavelength assumption is that the coherent-state labels vary smoothly from one lattice site to the next. Equivalently, there is a smooth unit spin field $\mathbf{S}(x, t)$ such that

$$\mathbf{S}_n(t) = \mathbf{S}(x_n, t), \quad \mathbf{S}(x + \ell, t) = \mathbf{S}(x, t). \quad (2.28)$$

For neighbouring sites this gives the Taylor expansion

$$\mathbf{S}_{n+1} = \mathbf{S} + \epsilon \partial_x \mathbf{S} + \frac{\epsilon^2}{2} \partial_x^2 \mathbf{S} + \mathcal{O}(\epsilon^3), \quad (2.29)$$

where the fields on the right-hand side are evaluated at (x_n, t) . The unit length constraint in (2.28) implies the useful kinematical identities

$$\mathbf{S}^\top \partial_x \mathbf{S} = 0, \quad \mathbf{S}^\top \partial_x^2 \mathbf{S} = -(\partial_x \mathbf{S})^\top (\partial_x \mathbf{S}). \quad (2.30)$$

For any smooth scalar density $f(x)$, the corresponding lattice sum obeys the usual Riemann-sum relation

$$\epsilon \sum_{n=1}^N f(x_n) \longrightarrow \int_0^\ell dx f(x). \quad (2.31)$$

The powers of ϵ that accompany the Hamiltonian, deformation parameters, and time coordinate are model dependent.

We now define the operation that maps local quantum operators to classical quantities. Let X_a be an operator acting on the auxiliary space $\mathcal{V}_a$ and on any finite set of physical lattice sites. Its *coherent-state lower symbol*, denoted by the superscript $\downarrow$ [10], is the auxiliary-space operator

$$X_a^\downarrow := (\mathbf{1}_{\mathcal{V}_a} \otimes \langle \Omega |) X_a (\mathbf{1}_{\mathcal{V}_a} \otimes | \Omega \rangle). \quad (2.32)$$

Only the physical sites on which X_a acts contribute to the matrix element; the remaining factors in (2.25) cancel by normalization. If X has no auxiliary-space indices, $X^\downarrow$ is a scalar function of the coherent-state labels. In particular, the lower symbol of the lattice Lax operator is

$$L_{a,n}^\downarrow(u) = \langle \Omega_n | L_{a,n}(u) | \Omega_n \rangle, \quad (2.33)$$

which is a 2×2 matrix acting on the auxiliary space.

The continuum spatial Lax matrix is extracted after the model-dependent normalization and spectral scaling have been fixed. Let λ denote the *continuum Lax spectral parameter*, to distinguish it from the quantum spectral parameter u defined in Section 2.1. For each model we choose a nonzero scalar normalization $c_L(\epsilon, \lambda)$ and a relation $u = u(\epsilon, \lambda)$ such that the normalized lattice operator

$$\hat{L}_{a,n}(\epsilon, \lambda) := c_L(\epsilon, \lambda) L_{a,n}(u(\epsilon, \lambda)) \quad (2.34)$$

has a regular long-wavelength expansion. When its lower symbol takes the form

$$\hat{L}_{a,n}^\downarrow(\epsilon, \lambda) = \mathbf{1}_{\mathcal{V}_a} + \epsilon U(x_n, t; \lambda) + \mathcal{O}(\epsilon^2), \quad (2.35)$$

we call the coefficient $U(x, t; \lambda)$ the *continuum spatial Lax matrix*. The choices of c_L , of the scaling $u(\epsilon, \lambda)$, and of any deformation parameters are part of the model-specific continuum limit; they are not fixed by the coherent-state construction alone.

The lower symbol of a single local operator is therefore straightforward. The next step is more delicate because the discrete zero-curvature equation (2.20) contains products of operators with overlapping physical support. We now determine how such products enter the continuum zero-curvature equation.

2.3 From the lattice Lax equation to the continuum zero-curvature equation

We now apply the lower-symbol operation to the finite-lattice equation (2.20). At this stage the lower symbol is taken at a fixed point of the coherent-state phase space. Applying it to the operator identity gives the exact relation

$$\begin{aligned} (\partial_{t_{\text{lat}}} L_{a,n}(u))^\downarrow &= (A_{a,n+1}(u) L_{a,n}(u))^\downarrow \\ &\quad - (L_{a,n}(u) A_{a,n}(u))^\downarrow. \end{aligned} \quad (2.36)$$

No identification with the derivative of a time-dependent coherent-state label is made here. For each model, the continuum Hamiltonian flow is derived independently and used to identify the continuum limit of the left-hand side with the time derivative of the classical spatial Lax matrix. The two products on the right-hand side cannot be simplified by applying the lower symbol to each factor separately, because the two factors in each product act on a common physical site. To keep track of this point without introducing any model-dependent formula, define the two auxiliary-space matrices

$$\mathcal{D}_{a,n}^{(+)}(u) := (A_{a,n+1}(u)L_{a,n}(u))^\downarrow - A_{a,n+1}^\downarrow(u)L_{a,n}^\downarrow(u), \quad (2.37)$$

$$\mathcal{D}_{a,n}^{(-)}(u) := (L_{a,n}(u)A_{a,n}(u))^\downarrow - L_{a,n}^\downarrow(u)A_{a,n}^\downarrow(u). \quad (2.38)$$

These matrices measure the failure of the lower symbol to preserve the operator product at the overlapping physical site. This nonmultiplicativity is standard in coherent-state symbol calculus and is commonly encoded by a Berezin star product [11].

With the definitions (2.37) and (2.38), the lower-symbol form of the lattice Lax equation is the exact identity

$$\begin{aligned} (\partial_{t_{\text{lat}}} L_{a,n}(u))^\downarrow &= A_{a,n+1}^\downarrow(u)L_{a,n}^\downarrow(u) - L_{a,n}^\downarrow(u)A_{a,n}^\downarrow(u) \\ &\quad + \mathcal{D}_{a,n}^{(+)}(u) - \mathcal{D}_{a,n}^{(-)}(u). \end{aligned} \quad (2.39)$$

For operators with disjoint physical support, the product coherent state (2.25) makes the corresponding difference vanish. The difference terms in (2.39) can survive because of the overlap described at the end of Section 2.1.

We next state the common continuum scaling used by the two models studied in this paper. The model-dependent spectral scaling and scalar normalization of the spatial Lax operator have already been represented abstractly by (2.34). We also use the freedom described below (2.20) to shift the finite-lattice time Lax operator by a scalar function times the identity. We denote the resulting operator by $\hat{A}_{a,n}$.

For both models, the continuum time coordinate t is related to the quantum lattice time t_{lat} by

$$t = \epsilon^2 t_{\text{lat}}. \quad (2.40)$$

Equation (2.40) fixes the normalization of the continuum time used in this paper. Since the scalar normalization of $\hat{L}_{a,n}$ and the scalar shift defining $\hat{A}_{a,n}$ are independent of t_{lat} and of the lattice site, the normalized operators obey the same discrete zero-curvature equation as $L_{a,n}$ and $A_{a,n}$. After the model-dependent spectral scaling has been imposed, the lower symbols of the normalized spatial and time Lax operators have the form

$$\hat{L}_{a,n}^\downarrow = \mathbf{1}_{\mathcal{V}_a} + \epsilon U(x_n, t; \lambda) + \mathcal{O}(\epsilon^2), \quad (2.41)$$

$$\hat{A}_{a,n}^\downarrow = \epsilon^2 V^\downarrow(x_n, t; \lambda) + \mathcal{O}(\epsilon^3). \quad (2.42)$$

Here $V^\downarrow(x, t; \lambda)$ denotes the continuum matrix obtained directly from the lower symbol of the finite-lattice time Lax operator. The scalings (2.40)–(2.42) are checked for each explicit model.

The part of (2.39) containing only products of lower symbols now has a standard continuum limit. Using $x_{n+1} = x_n + \epsilon$ together with (2.41) and (2.42), one finds

$$\begin{aligned} \hat{A}_{a,n+1}^\downarrow \hat{L}_{a,n}^\downarrow - \hat{L}_{a,n}^\downarrow \hat{A}_{a,n}^\downarrow \\ = \epsilon^3 \left(\partial_x V^\downarrow + [V^\downarrow, U] \right)_{(x_n, t; \lambda)} + \mathcal{O}(\epsilon^4). \end{aligned} \quad (2.43)$$

The right-hand side of the lattice equation therefore starts at order ϵ^3 . The coefficient of the lower symbol of the Heisenberg derivative at the same order is determined from the continuum Hamiltonian and its equations of motion for each model. This dynamical step is separate from the algebraic treatment of the operator products.

For the normalized operators, define the corresponding differences by

$$\widehat{\mathcal{D}}_{a,n}^{(+)} := (\widehat{A}_{a,n+1} \widehat{L}_{a,n})^\downarrow - \widehat{A}_{a,n+1}^\downarrow \widehat{L}_{a,n}^\downarrow, \quad (2.44)$$

$$\widehat{\mathcal{D}}_{a,n}^{(-)} := (\widehat{L}_{a,n} \widehat{A}_{a,n})^\downarrow - \widehat{L}_{a,n}^\downarrow \widehat{A}_{a,n}^\downarrow. \quad (2.45)$$

Their continuum contribution, when the limit exists, is

$$\mathcal{C}(x, t; \lambda) := \lim_{\epsilon \rightarrow 0} \frac{\widehat{\mathcal{D}}_{a,n}^{(+)} - \widehat{\mathcal{D}}_{a,n}^{(-)}}{\epsilon^3}, \quad x = x_n. \quad (2.46)$$

Once the model-specific Hamiltonian check identifies the continuum limit of the lower symbol of the Heisenberg derivative on the left-hand side of (2.39) with $\partial_t U$, the order- ϵ^3 equation becomes

$$\partial_t U = \partial_x V^\downarrow + [V^\downarrow, U] + \mathcal{C}. \quad (2.47)$$

The continuum zero-curvature equation is recovered if the matrix $\mathcal{C}$ can be written in the local form

$$\mathcal{C} = \partial_x \Delta V + [\Delta V, U] \quad (2.48)$$

for some auxiliary-space matrix $\Delta V(x, t; \lambda)$. In that case the continuum time Lax matrix

$$V := V^\downarrow + \Delta V \quad (2.49)$$

satisfies

$$\partial_t U - \partial_x V + [U, V] = 0. \quad (2.50)$$

Equations (2.46)–(2.50) formulate the continuum problem. The matrix ΔV is determined from the finite-lattice operator products for each model.

3 Class 5 model

3.1 Quantum Class 5 model and spatial Lax matrix

The name “Class 5” refers to one of the regular 4×4 difference-form solutions of the Yang–Baxter equation found in the classification of nearest-neighbour integrable spin- $\frac{1}{2}$ chains in Ref. [1]. Its continuum limit was subsequently studied in Ref. [2], where it was shown to be a non-Hermitian integrable deformation of the XXX spin chain whose long-wavelength limit is a deformed Landau–Lifshitz model. We use the same model, but choose from the outset a one-parameter representative convenient for the quantum-to-classical Lax calculation. This representative is related to the general Class 5 R -matrix by the spectral normalization and local basis transformation described in Ref. [2].

Let $\kappa \in \mathbb{C}$ denote the *lattice deformation parameter*. It corresponds to the parameter denoted by a in Ref. [2]. In the ordered two-spin basis

$$\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}, \quad (3.1)$$

the two-site Hamiltonian operator defining the Class 5 chain in our normalization is

$$h_5(\kappa) = \begin{pmatrix} 2 & 2\kappa & -2\kappa & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}. \quad (3.2)$$

Following the embedding convention of Section 2.1, the copy of h_5 on the bond joining sites n and $n + 1$ is denoted by $h_{5;n,n+1}(\kappa)$. The periodic quantum Hamiltonian is therefore

$$H_5^{\text{lat}}(\kappa) = \sum_{n=1}^N h_{5;n,n+1}(\kappa), \quad h_{5;N,N+1} \equiv h_{5;N,1}. \quad (3.3)$$

For $\kappa \neq 0$ this Hamiltonian is not Hermitian in the standard inner product. The undeformed value $\kappa = 0$ reduces, up to the normalization fixed in Section 2.1, to the spin- $\frac{1}{2}$ XXX chain.

The same normalization is generated by the regular Class 5 quantum R -matrix

$$R_5(u; \kappa) = \begin{pmatrix} 1 + 2u & 2\kappa u & -2\kappa u & 0 \\ 0 & 2u & 1 & 0 \\ 0 & 1 & 2u & 0 \\ 0 & 0 & 0 & 1 + 2u \end{pmatrix}, \quad (3.4)$$

where u is the quantum spectral parameter introduced in Section 2.1. It obeys the Yang–Baxter equation and the regularity condition

$$R_5(0; \kappa) = P, \quad (3.5)$$

and a direct differentiation gives

$$h_5(\kappa) = P \left. \frac{\partial R_5(u; \kappa)}{\partial u} \right|_{u=0}, \quad (3.6)$$

in agreement with the general relation (2.11). Equations (3.2) and (3.4) fix the model and the normalizations used in this section.

For the continuum expansion it is useful to isolate the deformation in a fixed two-site operator. Define

$$K := \begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \in \text{End}(\mathcal{H}_{\text{loc}} \otimes \mathcal{H}_{\text{loc}}). \quad (3.7)$$

With the permutation operator P defined in (2.9), the Class 5 data become

$$R_5(u; \kappa) = 2u\mathbf{1}_4 + P + 2\kappa uK, \quad h_5(\kappa) = 2P + 2\kappa K. \quad (3.8)$$

The operator K is nilpotent and satisfies

$$K^2 = 0, \quad PK = K, \quad KP = -K. \quad (3.9)$$

These elementary identities simplify the finite-lattice time Lax operator.

We now specify the long-wavelength scaling. The lattice deformation is sent to zero together with the lattice spacing ϵ , while the *continuum deformation coupling* α is held fixed:

$$\kappa = \epsilon\alpha. \quad (3.10)$$

This is the scaling used in the direct continuum construction of Ref. [2]; it places the deformation and the ordinary Landau–Lifshitz terms at the same order in the continuum action. For the Lax operator we simultaneously relate the quantum spectral parameter u to the continuum Lax spectral parameter λ by

$$u = \frac{\lambda}{\epsilon}, \quad \lambda \neq 0, \quad (3.11)$$

and use the common time scaling $t = \epsilon^2 t_{\text{lat}}$ introduced in (2.40).

For the fundamental spin- $\frac{1}{2}$ representation the lattice Lax operator is $L_{5;a,n}(u) := R_{5;a,n}(u; \kappa)$. We normalize it by its leading scalar part,

$$\widehat{L}_{5;a,n}(\epsilon, \lambda) := \frac{1}{2u} L_{5;a,n}(u), \quad (3.12)$$

where the scaling (3.10)–(3.11) is understood after the derivative-based quantum quantities have been formed. The normalized operator has the exact expansion

$$\widehat{L}_{5;a,n} = \mathbf{1}_{\mathcal{V}_a \otimes \mathcal{H}_n} + \epsilon L_{5;a,n}^{(1)}, \quad L_5^{(1)} = \frac{P}{2\lambda} + \alpha K. \quad (3.13)$$

Here $L_{5;a,n}^{(1)}$ denotes the copy of the fixed two-site operator $L_5^{(1)}$ on $\mathcal{V}_a \otimes \mathcal{H}_n$.

To evaluate its lower symbol, define the classical transverse spin combinations

$$S^+(x, t) := S^1(x, t) + iS^2(x, t), \quad S^-(x, t) := S^1(x, t) - iS^2(x, t). \quad (3.14)$$

Throughout this section, $S^\pm$ denotes these components of the classical unit spin field defined in (2.28). At the point $x = x_n$, the lower symbols of $P_{a,n}$ and $K_{a,n}$ are

$$P_{a,n}^\downarrow = \frac{1}{2} \begin{pmatrix} 1 + S^3 & S^- \\ S^+ & 1 - S^3 \end{pmatrix}, \quad (3.15)$$

$$K_{a,n}^\downarrow = \frac{1}{2} \begin{pmatrix} S^+ & -(1 + S^3) \\ 0 & 0 \end{pmatrix}. \quad (3.16)$$

Substituting these expressions into (3.13) gives

$$\widehat{L}_{5;a,n}^\downarrow = \mathbf{1}_{\mathcal{V}_a} + \epsilon U_5(x_n, t; \lambda), \quad (3.17)$$

where the *Class 5 continuum spatial Lax matrix* is

$$U_5(x, t; \lambda) = \frac{1}{4\lambda} \begin{pmatrix} 1 + S^3 + 2\alpha\lambda S^+ & S^- - 2\alpha\lambda(1 + S^3) \\ S^+ & 1 - S^3 \end{pmatrix}. \quad (3.18)$$

We use the complete $\mathfrak{gl}(2)$ matrix in (3.18).

Equation (3.18) completes the spatial part of the quantum-to-classical Lax map.

3.2 Time Lax matrix from the finite-lattice construction

We now take the long-wavelength limit of the finite-lattice time Lax operator. The starting point is the operator $A_{a,n}(u)$ defined in Eq. (2.19). Its addition by a scalar function of the quantum spectral parameter leaves the lattice zero-curvature equation unchanged. We use the representative

$$\widehat{A}_{5;a,n} := A_{5;a,n} + \frac{i}{u} \mathbf{1}, \quad (3.19)$$

where the identity acts on the auxiliary space and on the two physical sites $n-1, n$ supporting $A_{5;a,n}$. In terms of the normalized spatial operator $\widehat{L}_{5;a,n}$ in Eq. (3.12),

$$\widehat{A}_{5;a,n} = -i\widehat{L}_{5;a,n}^{-1} \left([h_{5;n-1,n}, \widehat{L}_{5;a,n}] + \partial_u \widehat{L}_{5;a,n} \right). \quad (3.20)$$

The derivative with respect to u is taken at fixed lattice deformation parameter κ . After imposing the continuum scalings (3.10) and (3.11),

$$\partial_u \widehat{L}_{5;a,n} = -\frac{P_{a,n}}{2u^2} = -\epsilon^2 \frac{P_{a,n}}{2\lambda^2}. \quad (3.21)$$

The Hamiltonian on the neighbouring physical sites has the expansion

$$h_{5;n-1,n} = h_{n-1,n}^{(0)} + \epsilon h_{n-1,n}^{(1)}, \quad h^{(0)} := 2P, \quad h^{(1)} := 2\alpha K. \quad (3.22)$$

Together with the exact relation $\widehat{L}_{5;a,n} = \mathbf{1} + \epsilon L_{5;a,n}^{(1)}$ in Eq. (3.13), Eq. (3.20) gives

$$\widehat{A}_{5;a,n} = \epsilon A_{a,n}^{(1)} + \epsilon^2 A_{a,n}^{(2)} + \mathcal{O}(\epsilon^3), \quad (3.23)$$

with

$$A_{a,n}^{(1)} = -i[h_{n-1,n}^{(0)}, L_{5;a,n}^{(1)}], \quad (3.24)$$

$$A_{a,n}^{(2)} = iL_{5;a,n}^{(1)}[h_{n-1,n}^{(0)}, L_{5;a,n}^{(1)}] - i[h_{n-1,n}^{(1)}, L_{5;a,n}^{(1)}] + \frac{i}{2\lambda^2} P_{a,n}. \quad (3.25)$$

Both coefficients act on the tensor product of the auxiliary space and the two neighbouring physical sites.

We first evaluate the lower symbol of Eq. (3.23). Let $\mathbf{S}_{n-1}$ and $\mathbf{S}_n$ be the coherent-state spin vectors on the sites $n-1$ and n . Direct contraction of Eq. (3.24) gives

$$(A_{a,n}^{(1)})^\downarrow = -2(\mathbf{S}_{n-1} \times \mathbf{S}_n)^i \frac{\partial U_5}{\partial S^i} \Big|_{\mathbf{S}=\mathbf{S}_n}. \quad (3.26)$$

Since

$$\mathbf{S}_{n-1} = \mathbf{S}(x_n, t) - \epsilon \partial_x \mathbf{S}(x_n, t) + \mathcal{O}(\epsilon^2), \quad (3.27)$$

we have

$$\mathbf{S}_{n-1} \times \mathbf{S}_n = \epsilon \mathbf{S} \times \partial_x \mathbf{S} + \mathcal{O}(\epsilon^2). \quad (3.28)$$

The term $A_{a,n}^{(2)}$ already carries the prefactor ϵ^2 in Eq. (3.23); its lower symbol is therefore needed only at coincident spin arguments at this order. Using the lower symbols in Eqs. (3.15) and (3.16), together with $\mathbf{S}^\top \mathbf{S} = 1$, gives

$$(A_{a,n}^{(2)})^\downarrow \Big|_{\mathbf{S}_{n-1}=\mathbf{S}_n=\mathbf{S}} = 4iU_5^2 - \frac{i}{2\lambda^2} \mathbf{1}_{\mathcal{V}_a}. \quad (3.29)$$

Equations (3.26)–(3.29) then yield

$$\widehat{A}_{5;a,n}^\downarrow = \epsilon^2 V_5^\downarrow(x_n, t; \lambda) + \mathcal{O}(\epsilon^3), \quad (3.30)$$

where

$$V_5^\downarrow = -2(\mathbf{S} \times \partial_x \mathbf{S})^i \frac{\partial U_5}{\partial S^i} + 4iU_5^2 - \frac{i}{2\lambda^2} \mathbf{1}_{\mathcal{V}_a}. \quad (3.31)$$

This is the continuum matrix obtained from the lower symbol of the finite-lattice time Lax operator itself.

The remaining contribution comes from the overlapping operator products in the discrete zero-curvature equation. Substituting Eqs. (3.23) and (3.13) into the differences defined in Eqs. (2.44) and (2.45), and keeping the first non-vanishing order, gives

$$\begin{aligned} & \widehat{\mathcal{D}}_{a,n}^{(+)} - \widehat{\mathcal{D}}_{a,n}^{(-)} \\ &= 2i\epsilon^3 \partial_x \left\{ \left[(L_5^{(1)})^2 \right]^\downarrow - \left[(L_5^{(1)})^\downarrow \right]^2 \right\} + \mathcal{O}(\epsilon^4). \end{aligned} \quad (3.32)$$

The algebra in Eq. (3.9) implies

$$(L_5^{(1)})^2 = \frac{1}{4\lambda^2} \mathbf{1}_4, \quad (3.33)$$

and Eq. (3.17) gives $(L_5^{(1)})^\downarrow = U_5$. Hence

$$\left[(L_5^{(1)})^2\right]^\downarrow - \left[(L_5^{(1)})^\downarrow\right]^2 = \frac{1}{4\lambda^2} \mathbf{1}_{\mathcal{V}_a} - U_5^2, \quad (3.34)$$

and the continuum contribution defined in Eq. (2.46) is

$$\mathcal{C}_5 = -2i \partial_x (U_5^2). \quad (3.35)$$

Equation (3.35) is absorbed into the time component by choosing

$$\Delta V_5 := -2i U_5^2 + \frac{i}{4\lambda^2} \mathbf{1}_{\mathcal{V}_a}. \quad (3.36)$$

It obeys

$$[\Delta V_5, U_5] = 0, \quad \partial_x \Delta V_5 + [\Delta V_5, U_5] = \mathcal{C}_5. \quad (3.37)$$

The scalar term in Eq. (3.36) fixes the representative; an x -independent scalar function of λ leaves the curvature unchanged. Combining Eqs. (3.31) and (3.36) gives

$$V_5 = -2(\mathbf{S} \times \partial_x \mathbf{S})^i \frac{\partial U_5}{\partial S^i} + 2i U_5^2 - \frac{i}{4\lambda^2} \mathbf{1}_{\mathcal{V}_a}. \quad (3.38)$$

Using the explicit spatial matrix in Eq. (3.18) and the unit-spin constraint, we write $S_x^a := \partial_x S^a$ for $a \in \{+, -, 3\}$. The time Lax matrix is

$$V_5(x, t; \lambda) = \begin{pmatrix} (V_5)_{11} & (V_5)_{12} \\ (V_5)_{21} & (V_5)_{22} \end{pmatrix}, \quad (3.39)$$

with

$$\begin{aligned} (V_5)_{11} = \frac{i}{4\lambda^2} & \left[S^3 + \alpha \lambda S^+ (1 + S^3) + 2\alpha^2 \lambda^2 (S^+)^2 \right. \\ & \left. + \lambda (S^- S_x^+ - S^+ S_x^-) + 4\alpha \lambda^2 (S^+ S_x^3 - S^3 S_x^+) \right], \end{aligned} \quad (3.40)$$

$$\begin{aligned} (V_5)_{12} = \frac{i}{4\lambda^2} & \left[S^- - \alpha \lambda (1 + S^3)^2 - 2\alpha^2 \lambda^2 S^+ (1 + S^3) \right. \\ & \left. + 2\lambda (S^3 S_x^- - S^- S_x^3) + 2\alpha \lambda^2 (S^+ S_x^- - S^- S_x^+) \right], \end{aligned} \quad (3.41)$$

$$(V_5)_{21} = \frac{i}{4\lambda^2} \left[S^+ + \alpha \lambda (S^+)^2 + 2\lambda (S^+ S_x^3 - S^3 S_x^+) \right], \quad (3.42)$$

$$(V_5)_{22} = \frac{i}{4\lambda^2} \left[-S^3 - \alpha \lambda S^+ (1 + S^3) + \lambda (S^+ S_x^- - S^- S_x^+) \right]. \quad (3.43)$$

The difference between Eqs. (3.31) and (3.38) is the contribution required by the overlapping operator products in the finite-lattice zero-curvature equation.

3.3 Zero-curvature equation and the Landau–Lifshitz equations of motion

We now determine the continuum Hamiltonian flow independently of the time Lax matrix derived in Section 3.2. This fixes the dynamical identification left open in Section 2.3 and provides a direct test of the corrected matrix V_5 .

It is convenient to introduce the constant antisymmetric matrix

$$M_5 := \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & -i \\ 1 & i & 0 \end{pmatrix}, \quad M_5^\top = -M_5. \quad (3.44)$$

The lower symbol of a single Class 5 bond Hamiltonian, evaluated on the coherent states at the neighbouring points x and $x + \epsilon$, has the long-wavelength expansion

$$h_5^\downarrow(\mathbf{S}(x, t), \mathbf{S}(x + \epsilon, t)) = 2 + \epsilon^2 \mathcal{H}_5(x, t) + \mathcal{O}(\epsilon^3), \quad (3.45)$$

where

$$\mathcal{H}_5 = -\frac{1}{2} \partial_x \mathbf{S}^\top \partial_x \mathbf{S} + \frac{\alpha}{2} \left(\mathbf{S}^\top M_5 \partial_x \mathbf{S} + \partial_x S^+ \right). \quad (3.46)$$

The last term is a total spatial derivative. With the periodic boundary condition in (2.28), the continuum Hamiltonian obtained from the lattice Hamiltonian is therefore

$$\begin{aligned} H_5^{\text{cl}} &:= \lim_{\epsilon \rightarrow 0} \frac{(H_5^{\text{lat}})^\downarrow - 2N}{\epsilon} \\ &= \int_0^\ell dx \left[-\frac{1}{2} \partial_x \mathbf{S}^\top \partial_x \mathbf{S} + \frac{\alpha}{2} \mathbf{S}^\top M_5 \partial_x \mathbf{S} \right]. \end{aligned} \quad (3.47)$$

The signs and the overall normalization in (3.47) follow from the lattice Hamiltonian (3.3), the Heisenberg equation (2.18), and the continuum time convention $t = \epsilon^2 t_{\text{lat}}$.

For the unit spin field normalized in (2.28), we use the continuum Poisson bracket

$$\{S^i(x), S^j(y)\}_5 = 2 \varepsilon_{ijk} S^k(x) \delta(x - y), \quad \varepsilon_{123} = 1, \quad (3.48)$$

where $\delta(x - y)$ is the Dirac delta function. The corresponding Hamiltonian flow is

$$\partial_t S^i(x) = \{S^i(x), H_5^{\text{cl}}\}_5. \quad (3.49)$$

Using the bilinear cross product $(\mathbf{A} \times \mathbf{B})^i = \varepsilon_{ijk} A^j B^k$, the equations of motion take the compact form

$$\partial_t \mathbf{S} = -2 \mathbf{S} \times \left(\partial_x^2 \mathbf{S} + \alpha M_5 \partial_x \mathbf{S} \right). \quad (3.50)$$

The right-hand side is tangent to the unit-spin constraint, so $\partial_t(\mathbf{S}^\top \mathbf{S}) = 0$.

For comparison with the matrix entries of the Lax pair, the same equations can be written in terms of the transverse components $S^\pm$ defined in (3.14). Using $\mathbf{S}^\top \partial_x \mathbf{S} = 0$, one obtains

$$\partial_t S^+ = 2i \left(S^+ \partial_x^2 S^3 - S^3 \partial_x^2 S^+ + \alpha S^+ \partial_x S^+ \right), \quad (3.51)$$

$$\partial_t S^- = 2i \left(S^3 \partial_x^2 S^- - S^- \partial_x^2 S^3 + \alpha S^+ \partial_x S^- \right), \quad (3.52)$$

$$\partial_t S^3 = i \left(S^- \partial_x^2 S^+ - S^+ \partial_x^2 S^- + 2\alpha S^+ \partial_x S^3 \right). \quad (3.53)$$

A direct expansion of the lower symbol of the lattice Heisenberg derivative then gives

$$\left(\partial_{t_{\text{lat}}} \widehat{L}_{5;a,n} \right)^\downarrow = \epsilon^3 \partial_t U_5(x_n, t; \lambda) + \mathcal{O}(\epsilon^4), \quad (3.54)$$

with $\partial_t \mathbf{S}$ given by (3.50). Equation (3.54) therefore supplies the dynamical identification required on the left-hand side of the continuum equation (2.47).

We finally test the relation between the corrected Lax pair and the equations of motion without imposing the equations of motion in the curvature. Define the three equation-of-motion residuals

$$E_+ := \partial_t S^+ - 2i \left(S^+ \partial_x^2 S^3 - S^3 \partial_x^2 S^+ + \alpha S^+ \partial_x S^+ \right), \quad (3.55)$$

$$E_- := \partial_t S^- - 2i \left(S^3 \partial_x^2 S^- - S^- \partial_x^2 S^3 + \alpha S^+ \partial_x S^- \right), \quad (3.56)$$

$$E_3 := \partial_t S^3 - i \left(S^- \partial_x^2 S^+ - S^+ \partial_x^2 S^- + 2\alpha S^+ \partial_x S^3 \right). \quad (3.57)$$

The curvature of the matrices U_5 and V_5 in (3.18) and (3.39) is

$$F_{tx}^{(5)} := \partial_t U_5 - \partial_x V_5 + [U_5, V_5]. \quad (3.58)$$

Using only the unit-spin constraint and its spatial derivatives, the curvature factorizes as

$$F_{tx}^{(5)} = \frac{1}{4\lambda} \begin{pmatrix} E_3 + 2\alpha\lambda E_+ & E_- - 2\alpha\lambda E_3 \\ E_+ & -E_3 \end{pmatrix}. \quad (3.59)$$

For $\lambda \neq 0$, the residuals are recovered from the curvature entries by

$$E_+ = 4\lambda F_{21}^{(5)}, \quad E_3 = -4\lambda F_{22}^{(5)}, \quad E_- = 4\lambda (F_{12}^{(5)} - 2\alpha\lambda F_{22}^{(5)}). \quad (3.60)$$

Hence the zero-curvature condition $F_{tx}^{(5)} = 0$ is equivalent to the three continuum equations of motion (3.51)–(3.53).

To compare with the direct continuum construction of Ref. [2], we translate our conventions by

$$S^2 \mapsto -S^2, \quad \alpha \mapsto -\alpha, \quad t \mapsto 2t. \quad (3.61)$$

With these substitutions, Eq. (3.50) takes the form of the Class 5 continuum equation used in Ref. [2]. The reflection of the second spin axis conjugates M_5 by $\text{diag}(1, -1, 1)$, yielding the deformation matrix used there.

3.4 Comparison with the classical r -matrix construction

The time Lax matrix in Eq. (3.39) was obtained by starting from the finite-lattice time operator. We now construct the time component independently from the classical r -matrix and monodromy of the continuum model. This provides a check that does not use the finite-lattice time operator $A_{a,n}$.

To keep the two roles of the spectral parameter separate, let $\zeta \in \mathbb{C}$ denote the *generating spectral parameter* used to expand the transfer matrix, while λ continues to denote the Lax spectral parameter of the final pair (U_5, V_5) . Define the rank-one coherent-state projector

$$\rho(x, t) := \frac{1}{2} \begin{pmatrix} 1 + S^3 & S^- \\ S^+ & 1 - S^3 \end{pmatrix}, \quad \rho^2 = \rho, \quad \text{tr } \rho = 1, \quad (3.62)$$

and the deformation matrix in the auxiliary space

$$K_5^{\text{cl}}(x, t) := \frac{1}{2} \begin{pmatrix} S^+ & -(1 + S^3) \\ 0 & 0 \end{pmatrix}. \quad (3.63)$$

These are precisely the lower symbols in Eqs. (3.15) and (3.16). The spatial Lax matrix therefore admits the form

$$U_5(x, t; \lambda) = \frac{\rho(x, t)}{2\lambda} + \alpha K_5^{\text{cl}}(x, t). \quad (3.64)$$

We first fix the classical r -matrix in the conventions of this paper. Let P_{12} be the permutation operator on two copies of the two-dimensional auxiliary space. We also define

$$P_{\uparrow} := \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad E_+ := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad (3.65)$$

and the constant Jordanian tensor

$$C_{12} := P_{\uparrow} \otimes E_+ - E_+ \otimes P_{\uparrow}. \quad (3.66)$$

A direct calculation with the spin Poisson bracket (3.48) gives

$$r_{12}^{(5)}(\zeta, \lambda) = \frac{i}{2(\zeta - \lambda)} P_{12} + i\alpha C_{12}, \quad (3.67)$$

for which

$$\{U_{5,1}(x; \zeta), U_{5,2}(y; \lambda)\}_5 = [r_{12}^{(5)}(\zeta, \lambda), U_{5,1}(x; \zeta) + U_{5,2}(y; \lambda)] \delta(x - y). \quad (3.68)$$

Here $U_{5,1} := U_5 \otimes \mathbf{1}_2$ and $U_{5,2} := \mathbf{1}_2 \otimes U_5$ act on the first and second auxiliary spaces, respectively. The rational part of Eq. (3.67) is the usual isotropic Landau–Lifshitz contribution, whereas the constant term contains the Class 5 deformation. This structure is consistent with the classical r -matrix obtained from the Class 5 quantum model in Ref. [2].

For completeness, define the classical monodromy matrix by

$$T(x, y; \zeta) := \mathcal{P} \exp \left(\int_y^x U_5(s, t; \zeta) ds \right), \quad (3.69)$$

and the associated transfer matrix on the periodic interval of length ℓ by

$$\tau(\zeta) := \text{tr } T(\ell, 0; \zeta). \quad (3.70)$$

For a linear classical r -matrix algebra of the form (3.68), the standard monodromy construction gives the generating time matrix

$$\mathbb{V}_2(x; \zeta, \lambda) = \frac{1}{\tau(\zeta)} \text{tr}_1 \left[T_1(\ell, x; \zeta) r_{12}^{(5)}(\zeta, \lambda) T_1(x, 0; \zeta) \right], \quad (3.71)$$

where the trace is taken over the first auxiliary space [8, 7].

The pole of Eq. (3.64) at small ζ is proportional to the rank-one projector ρ . The local expansion of the monodromy is therefore governed by a rank-one projector $\Pi(x; \zeta)$ satisfying

$$\Pi^2 = \Pi, \quad \text{tr } \Pi = 1, \quad \partial_x \Pi = [U_5(x; \zeta), \Pi], \quad (3.72)$$

with the branch

$$\Pi(x; \zeta) = \rho(x) + \zeta \Pi_1(x) + \mathcal{O}(\zeta^2). \quad (3.73)$$

Expanding Eq. (3.72) to first order fixes the coefficient uniquely as

$$\Pi_1 = 2[\rho, \partial_x \rho] - 2\alpha[\rho, [K_5^{\text{cl}}, \rho]]. \quad (3.74)$$

Thus the first correction contains both the ordinary spatial-gradient term and the deformation term.

Using Eq. (3.73) in the generating matrix (3.71), write

$$\mathbb{V}_2(x; \zeta, \lambda) = \mathbb{V}^{(0)}(x; \lambda) + \zeta \mathbb{V}^{(1)}(x; \lambda) + \mathcal{O}(\zeta^2). \quad (3.75)$$

The coefficient relevant for the Class 5 Hamiltonian flow is

$$\begin{aligned} \mathbb{V}^{(1)} = & -\frac{i}{2\lambda} \left(\Pi_1 + \frac{\rho}{\lambda} \right) \\ & + i\alpha [\text{tr}(\Pi_1 P_{\uparrow}) E_+ - \text{tr}(\Pi_1 E_+) P_{\uparrow}]. \end{aligned} \quad (3.76)$$

The standard monodromy formula is conventionally written with the Hamiltonian in the first argument of the Poisson bracket. Our Hamiltonian evolution in Eq. (3.49) uses the opposite

order. With this convention, the time Lax matrix obtained from the classical r -matrix route is therefore

$$V_{5,r}(x; \lambda) := -\mathbb{V}^{(1)}(x; \lambda). \quad (3.77)$$

Substituting Eq. (3.74) into Eq. (3.76) and using only the unit spin constraint gives

$$V_{5,r}(x; \lambda) - V_5(x; \lambda) = \frac{i}{4\lambda^2} \mathbf{1}_2. \quad (3.78)$$

The matrix on the right-hand side is independent of the spin field and of the spatial coordinate. Its spatial derivative vanishes and it commutes with U_5 . Hence $V_{5,r}$ and the finite-lattice result V_5 generate the same zero-curvature equation. The two constructions thus recover the same Class 5 time Lax component up to the scalar freedom already encountered in Eq. (3.36).

4 Class 6 model

We now apply the finite-lattice construction to the Class 6 chain. The name refers to the regular difference-form solution classified in Ref. [1], subsequently identified with the eleven-vertex model in Ref. [3]. Its associated Landau–Lifshitz hierarchy has been studied in Ref. [4]. Here we use the quantum-chain normalization of Appendix A of Ref. [2] to construct the continuum Lax matrices. The spatial expansion contains a nonzero second-order operator coefficient, which must be retained when taking the continuum limit of the time component.

4.1 Quantum Class 6 model and continuum limit

Let $\kappa_6 \in \mathbb{C}$ denote the *Class 6 lattice deformation parameter*. We use the ordered two-spin basis $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$ and fix the coefficient of the permutation interaction to be 2. In this normalization the local Hamiltonian is

$$h_6(\kappa_6) = \begin{pmatrix} 2 & \kappa_6 & \kappa_6 & 0 \\ 0 & 0 & 2 & -\kappa_6 \\ 0 & 2 & 0 & -\kappa_6 \\ 0 & 0 & 0 & 2 \end{pmatrix} = 2P + \kappa_6 K_6, \quad (4.1)$$

where P is the permutation operator in Eq. (2.9), and the two-site deformation operator K_6 is defined by

$$K_6 := \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (4.2)$$

Its algebra with P is

$$PK_6 = K_6P = K_6, \quad K_6^3 = 0, \quad K_6^2 = \begin{pmatrix} 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (4.3)$$

For the periodic chain of N sites, the quantum Hamiltonian is

$$H_6^{\text{lat}}(\kappa_6) = \sum_{n=1}^N h_{6;n,n+1}(\kappa_6), \quad h_{6;N,N+1} \equiv h_{6;N,1}. \quad (4.4)$$

The Hamiltonian is non-Hermitian for $\kappa_6 \neq 0$ in the standard inner product. At $\kappa_6 = 0$ it reduces to the spin- $\frac{1}{2}$ XXX chain with the same exchange normalization as in Section 3.1.

Using the quantum spectral parameter u of Section 2.1, the regular R -matrix generating Eq. (4.1) is

$$R_6(u; \kappa_6) = 2u\mathbf{1}_4 + P + \kappa_6 u(1+2u)K_6 + \frac{\kappa_6^2 u^2 (1+2u)^2}{2} K_6^2 \quad (4.5)$$

$$= \begin{pmatrix} 1+2u & \kappa_6 u(1+2u) & \kappa_6 u(1+2u) & -\kappa_6^2 u^2 (1+2u)^2 \\ 0 & 2u & 1 & -\kappa_6 u(1+2u) \\ 0 & 1 & 2u & -\kappa_6 u(1+2u) \\ 0 & 0 & 0 & 1+2u \end{pmatrix}. \quad (4.6)$$

This is Eq. (A.1) of Ref. [2] with the exchange normalization fixed above. It satisfies the Yang–Baxter equation and

$$R_6(0; \kappa_6) = P, \quad P \partial_u R_6(u; \kappa_6)|_{u=0} = h_6(\kappa_6). \quad (4.7)$$

All quantum spectral derivatives are taken at fixed κ_6 . The identities in Eq. (4.3) also give

$$R_6(u; \kappa_6) R_6(-u; \kappa_6) = (1 - 4u^2) \mathbf{1}_4. \quad (4.8)$$

Thus the inverse required by the finite-lattice time operator exists for $u \neq \pm \frac{1}{2}$. Equation (4.8) is an algebraic inversion relation; it contains no Hermitian conjugation.

To fix the continuum scaling, consider the lower symbol of a bond Hamiltonian. The coherent states and the smooth unit spin field $\mathbf{S}(x, t)$ are those of Section 2.2. We write the transverse components as

$$S^\pm := S^1 \pm iS^2, \quad S_n^\pm := S^\pm(x_n, t). \quad (4.9)$$

The algebraic expressions below also extend to complex spin components subject to $\mathbf{S}^\top \mathbf{S} = 1$, with the transpose defining a bilinear product. On this complexified spin space, S^+ and S^- need not be complex conjugates. Direct evaluation of Eq. (4.1) gives

$$h_{6;n,n+1}^\downarrow = 1 + \mathbf{S}_n^\top \mathbf{S}_{n+1} + \frac{\kappa_6}{2} (S_n^+ S_{n+1}^3 + S_n^3 S_{n+1}^+). \quad (4.10)$$

After subtracting the constant 2, the exchange term starts at order ϵ^2 by Eq. (2.30), whereas the deformation term tends to $\kappa_6 S^+ S^3$ for coincident spins. Both terms therefore survive in the same continuum limit if the *Class 6 continuum deformation coupling* α_6 and the continuum spectral parameter $\lambda \neq 0$ are held fixed under

$$\kappa_6 = \epsilon^2 \alpha_6, \quad u = \frac{\lambda}{\epsilon}, \quad t = \epsilon^2 t_{\text{lat}}, \quad \lambda \neq 0. \quad (4.11)$$

Here ϵ is the lattice spacing, u and λ are the quantum and continuum spectral parameters, and t_{lat} and t are the lattice and continuum times. The deformation scaling agrees with Appendix A.1 of Ref. [2]; the spectral scaling ensures that the normalized spatial Lax operator approaches the identity. If lattice parameters are dimensionless, λ has the dimension of length and α_6 has the dimension of inverse length squared.

In the fundamental representation, define $L_{6;a,n}(u) := R_{6;a,n}(u; \kappa_6)$ on $\mathcal{V}_a \otimes \mathcal{H}_n$ and normalize it as

$$\widehat{L}_{6;a,n}(\epsilon, \lambda) := \frac{1}{2u} L_{6;a,n}(u). \quad (4.12)$$

Substituting Eq. (4.11) into Eq. (4.5) yields the exact finite expansion

$$\widehat{L}_{6;a,n} = \mathbf{1}_{\mathcal{V}_a \otimes \mathcal{H}_n} + \epsilon L_{6;a,n}^{(1)} + \epsilon^2 L_{6;a,n}^{(2)} + \epsilon^3 L_{6;a,n}^{(3)}. \quad (4.13)$$

The superscript labels the power of ϵ , and each coefficient is the copy on $\mathcal{V}_a \otimes \mathcal{H}_n$ of the two-site operator

$$L_6^{(1)} = \frac{P}{2\lambda} + \alpha_6 \lambda K_6 + \alpha_6^2 \lambda^3 K_6^2, \quad (4.14)$$

$$L_6^{(2)} = \frac{\alpha_6}{2} K_6 + \alpha_6^2 \lambda^2 K_6^2, \quad L_6^{(3)} = \frac{\alpha_6^2 \lambda}{4} K_6^2. \quad (4.15)$$

In particular, the operator square of the first coefficient satisfies

$$(L_6^{(1)})^2 = \frac{\mathbf{1}_4}{4\lambda^2} + 2L_6^{(2)}. \quad (4.16)$$

The second coefficient is therefore already present in products of the leading spatial coefficients. It will enter the coherent-state product correction in the time Lax equation.

For the lower-symbol evaluation at $x = x_n$, the permutation gives

$$P_{a,n}^\downarrow = \frac{1}{2} \begin{pmatrix} 1 + S^3 & S^- \\ S^+ & 1 - S^3 \end{pmatrix}, \quad (4.17)$$

while the deformation operators give

$$K_{6;a,n}^\downarrow = \begin{pmatrix} S^+/2 & S^3 \\ 0 & -S^+/2 \end{pmatrix}, \quad (K_{6;a,n}^2)^\downarrow = \begin{pmatrix} 0 & -S^+ \\ 0 & 0 \end{pmatrix}. \quad (4.18)$$

The second matrix is obtained by squaring the quantum operator before taking its lower symbol. Using these matrices in Eq. (4.14), the continuum spatial Lax matrix is

$$\begin{aligned} U_6(x, t; \lambda) &:= (L_{6;a,n}^{(1)})^\downarrow|_{x_n=x} \\ &= \frac{1}{4\lambda} \begin{pmatrix} 1 + S^3 + 2\alpha_6 \lambda^2 S^+ & S^- + 4\alpha_6 \lambda^2 S^3 - 4\alpha_6^2 \lambda^4 S^+ \\ S^+ & 1 - S^3 - 2\alpha_6 \lambda^2 S^+ \end{pmatrix}. \end{aligned} \quad (4.19)$$

It obeys

$$\widehat{L}_{6;a,n}^\downarrow = \mathbf{1}_{\mathcal{V}_a} + \epsilon U_6(x_n, t; \lambda) + \mathcal{O}(\epsilon^2), \quad \text{tr } U_6 = \frac{1}{2\lambda}. \quad (4.20)$$

We retain the complete $\mathfrak{gl}(2)$ matrix in the finite-lattice limit and denote its traceless part by

$$U_{6,0} := U_6 - \frac{1}{4\lambda} \mathbf{1}_2. \quad (4.21)$$

At $\alpha_6 = 0$, Eq. (4.19) reduces to $(\mathbf{1}_2 + S^i \sigma^i)/(4\lambda)$, the spatial Lax matrix of the isotropic Landau–Lifshitz model in the normalization used here. For the deformed model, Eqs. (4.13)–(4.16) provide the operator coefficients needed to construct the time component.

4.2	Time Lax matrix from the finite-lattice construction
4.3	Zero-curvature equation and equations of motion
5	Comparison of Class 5 and Class 6
5.1	Operator-product corrections in the coherent-state limit
5.2	Common structure and model-dependent terms
5.3	Scope of the construction
6	Conclusions
A	Conventions and normalisations
B	Details of the Class 5 continuum expansion
C	Details of the Class 6 continuum expansion
D	Classical r -matrix and monodromy construction for Class 5
E	Symbolic and finite-lattice checks

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